PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-5

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TOPICS TO BE COVERED

1. Comparing Electrostatic force with Gravitational force
2. Motion of a charged particle in a uniform electric field
3. Electric field of a continuous charge distribution
4. Volume, surface and linear charge distribution

Recap

1. Electric Field → $E = \frac{F}{q}$, $E = \frac{kq}{r^2}$ → N/C / V/m.
2. Electric Field due to a point charge
3. Electric field due to a group of charges → Superposition Principle.

Parishram 2.0 students when I extend 1.5 hrs class to more than 2 hrs

Students:



QUESTION

CUET-2022

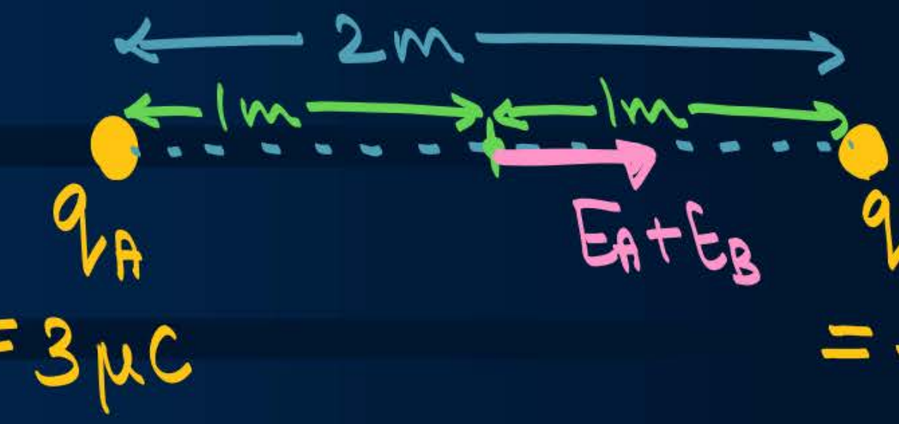
Two point charges $q_A = 3\ \mu C$ and $q_B = -3\ \mu C$ are located 2 m apart in vacuum. The electric field at midpoint of the line joining the two charges is

A 5.4×10^4 N/C

B 1.35×10^4 N/C

C 2.7×10^4 N/C

D Zero



$$E_A + E_B = \frac{k(3\mu C)}{(1)^2} + \frac{k(3\mu C)}{(1)^2} = \frac{2\times(3\times10^{-6})\times9\times10^{9}}{1}$$

$$= 5.4\times10^{4}\ N/C$$

QUESTION

CUET

The magnitude of electric force F exerted by a charged particle on another charged particle is 2.4×10^{-8} N. If the distance between them becomes half of its previous value, then the magnitude of the exerted force will be

A 1.2×10^{-8} N

B 4.8×10^{-8} N

C 9.6×10^{-8} N

D 19.2×10^{-8} N

$$F = \frac{kq_1q_2}{r^2} = 2.4 \times 10^{-8}\ N$$

$$F' = \frac{kq_1q_2}{\left(\frac{r}{2}\right)^2} = 4\frac{kq_1q_2}{r^2}$$

$$\Rightarrow F' = 4 \times 2.4 \times 10^{-8}\ N = \boxed{9.6 \times 10^{-8}\ N}$$



Electric field is defined as :-

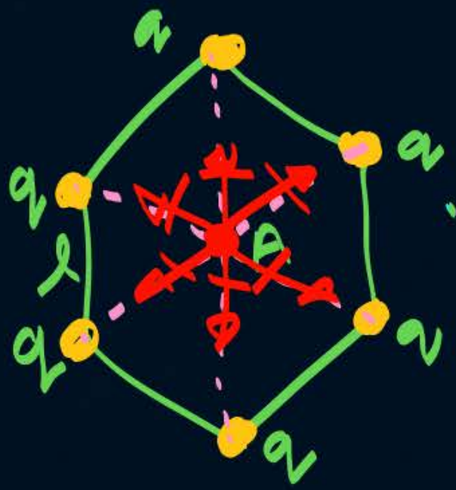
(A) $\frac{kQq}{r^2}$

(B) $\frac{kQ}{r}$

(C) $\frac{kQq}{r}$

(D) $\frac{kQ}{r^2}$

Que→ Regular Hexagon:



what will be the $\vec{E}$ at the midpoint A?

DADA method

A) $\frac{kq}{2\sqrt{2}l^2}$

B) $\frac{kq}{\sqrt{3}l^2}$

C) Zero

D) $\frac{kq}{2l^2}$

Comparing Electrostatic force with Gravitational force

Gravitational force :-

$$F_g = \frac{GmM}{r^2} \ N$$

G = Gravitational Constant

Electrostatic force :-

$$F_e = \frac{kqQ}{r^2} \ N$$

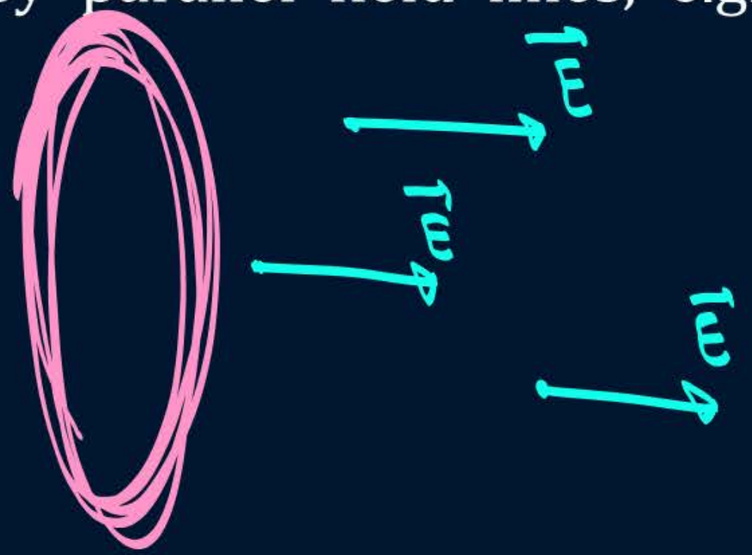
k = Electrostatic Constant

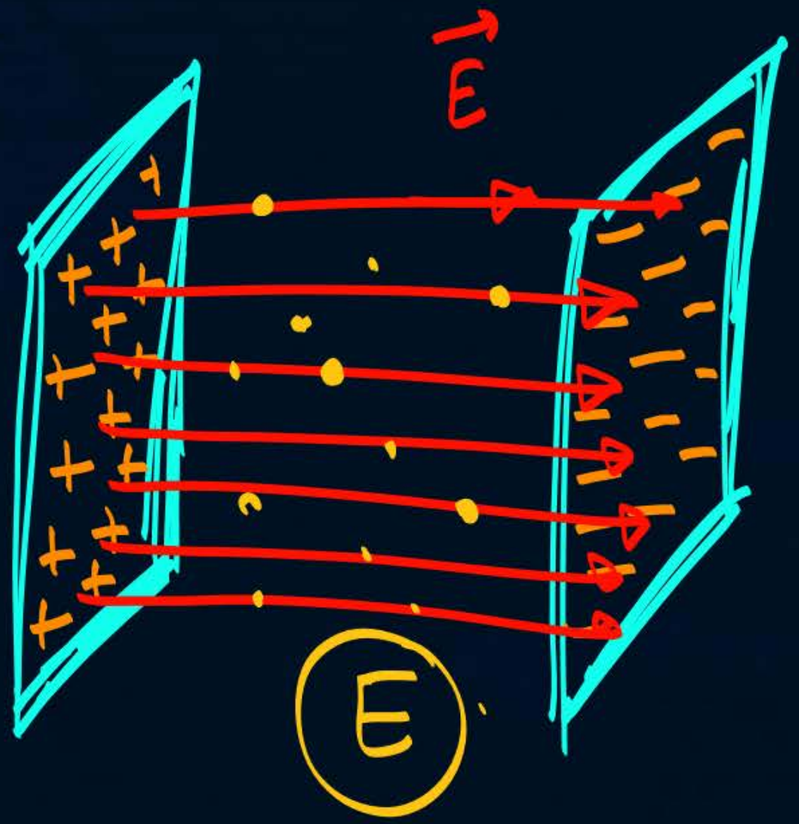


Motion of a charged particle in a uniform electric field

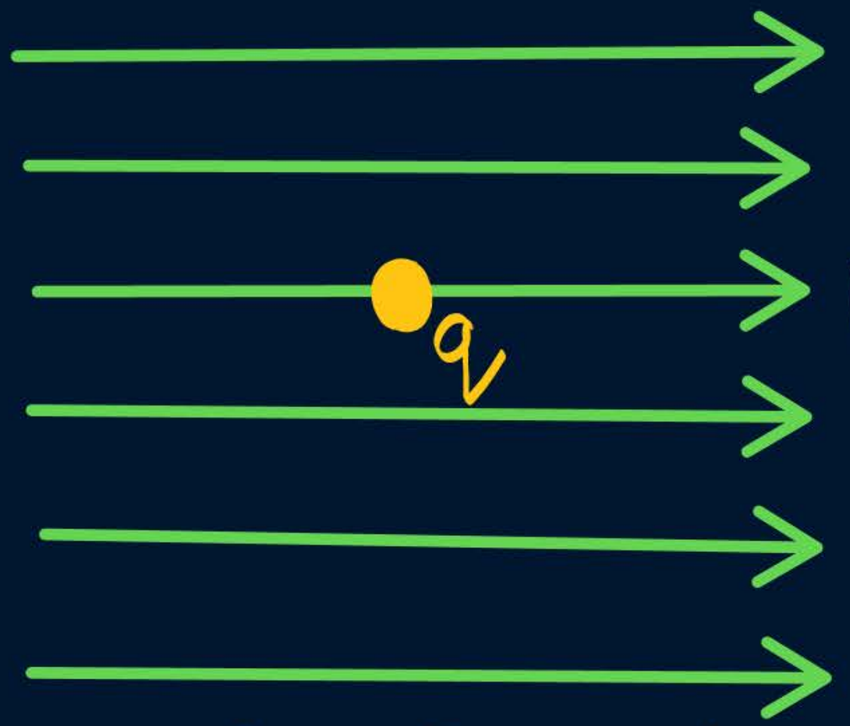
➢ Uniform Electric Field

- A uniform electric field means the field has the same magnitude and direction at every point.
- It is often represented by parallel field lines, e.g., between two oppositely charged parallel plates.




$\vec{E}$
E

Motion of a charged particle in a uniform electric field



Uniform Electric field.

$F = qE$

$\Rightarrow ma = qE$

$\Rightarrow a = \frac{qE}{m}$ ⇒ Constant acceleration.

→ Acceleration of the charged particle in a Uniform Electric field.

$\vec{F} = m\vec{a}$

चेला

गुरु

if the charge is initially at rest, then

initial velocity = $u = 0\ m/s$

we know that, $v = u + at$

$\Rightarrow v = at$

$\Rightarrow \boxed{v = \frac{qE}{m}t}$

velocity of the charged particle after time 't' in the Uniform Electric field.

$$S = ut + \frac{1}{2}at^2$$

$\rightarrow u = 0$ m/s

$\rightarrow a = \frac{qE}{m}$

$$x = (0)t + \frac{1}{2}\left(\frac{qE}{m}\right)t^2$$

$$\Rightarrow \boxed{x = \frac{1}{2}\frac{qE}{m}t^2}$$

$\rightarrow$ Displacement of the charged particle after time t in the direction of Uniform Electric Field.

QUESTION

A particle of charge $q = 2 \times 10^{-6}$ C and mass m = 1 kg is placed in a uniform electric field E = 3000 N/C. What is the acceleration of the particle?

A. $3\ m/s^2$

B. $6\ m/s^2$

C. $0.006\ m/s^2$

D. $0.003\ m/s^2$

$$a = \frac{qE}{m}$$

$$a = \frac{(2 \times 10^{-6})(3000)}{(1)}$$

$$a = 0.006\ m/s^2$$



QUESTION

A particle of mass 5 g and charge +2 μC is released from rest in a uniform electric field of $E = 2 \times 10^4$ N/C. What is its kinetic energy after moving 0.6 m in the direction of the field?

A 0.024 J

B 0.012 J

C 0.048 J

D 0.006 J

$\frac{1}{2}mv^2$

Work-Energy theorem.

Work = $\Delta K.E.$

$\Rightarrow F.d = (K.E.)_f - \left(\frac{1}{2}mv^2\right)$ (v = 0, so this term is $K.E._i$)

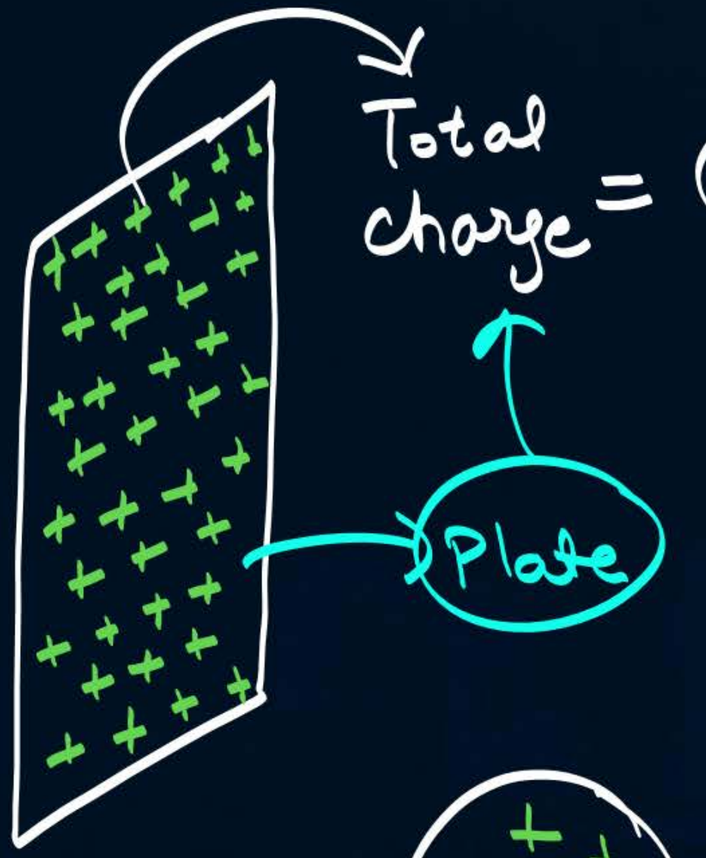
$\Rightarrow (qE)d = (K.E.)_f$

$\Rightarrow 2\times10^{-6} \times 2\times10^{4} \times 6\times10^{-1} = 24\times10^{-3}$ J

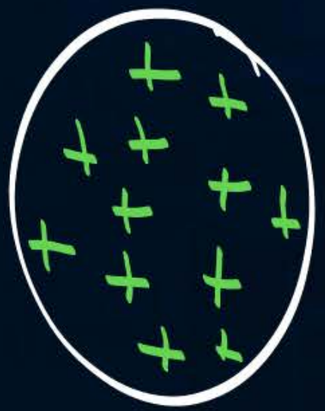
$\therefore$




Q
q
F=qE
E


Total charge = Q
Plate

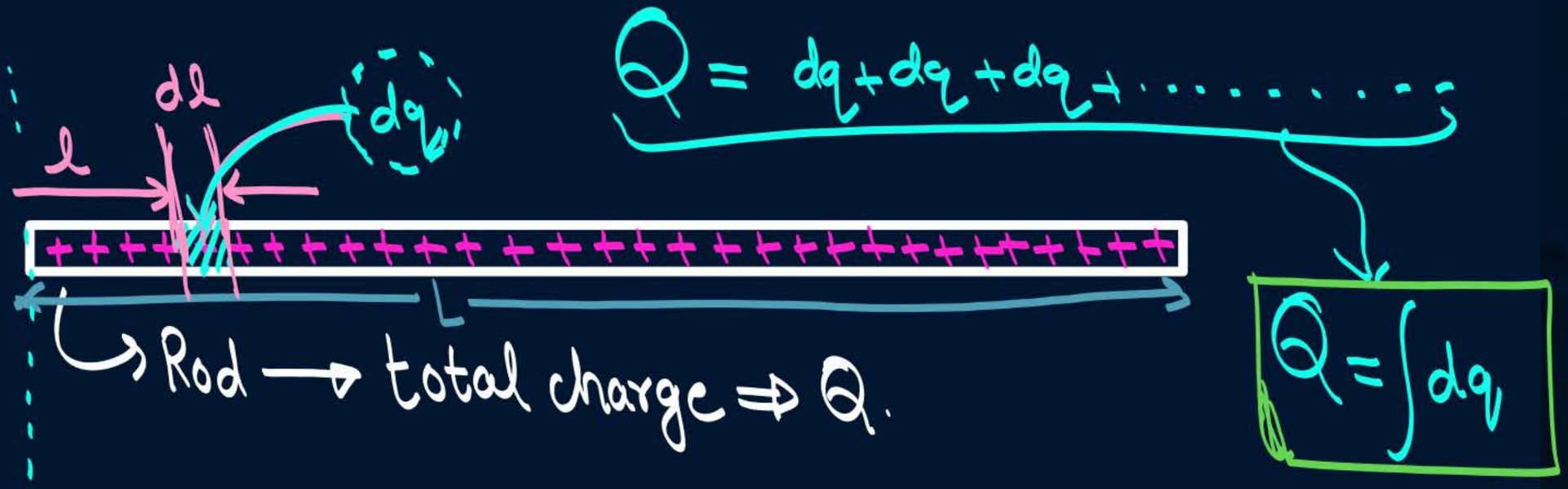

Q





Electric Field of a Continuous Charge Distribution

- The electric field at a point due to a continuous distribution is the vector sum (integral) of the fields due to all infinitesimal charge elements dq in the distribution.



Point charge

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

Electric field due to the point charge.

$$E = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r^2}$$

→ Electric field due to continuous charge distribution.

Electric charge Density

Density = ~~mass / Volume~~

Charge Density = $\frac{Q}{L} = \lambda$

→ linear charge density.

Linear Charge Density

Linear charge density (λ) is defined as the charge per unit length.

Lambda

$$\lambda = \frac{dq}{dl}$$

$$\lambda = \frac{Q}{L} = \frac{dq}{dl}$$

$$dq = \frac{Q}{L}\,dl$$

λ= constant (for uniform distribution)

= variable (for non uniform distribution)

where dq is the charge on an infinitesimal length dl.

Unit of λ is coulomb/meter (C/m)

Examples: Straight charged wire, charged ring

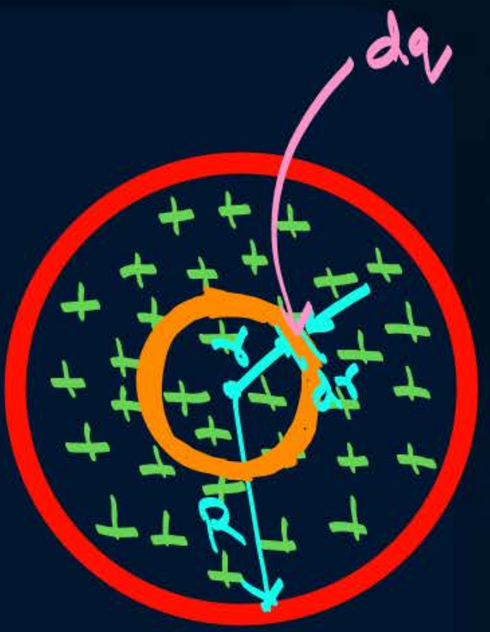
Surface charge density

Surface charge density (σ) is defined as the charge per unit area.

$$\sigma = \frac{dq}{ds}$$

Sigma

$\sigma = \frac{Q}{A}$ C / m^2



σ = constant (uniform distribution)

= variable (non uniform distribution)

where dq is the charge on an infinitesimal surface area ds. Unit of (σ) is coulomb/meter^2 (C/m^2).

Examples: Plane sheet of charge, conducting sphere.

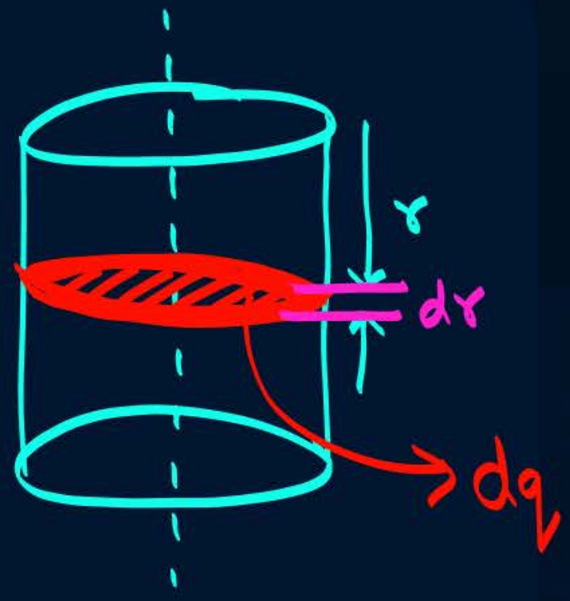
Volume Charge density

Volume charge density (ρ) is defined as charge per unit volume.

$$\rho = \frac{dq}{dV}$$

rho

$$\rho = \frac{Q}{V}$$



ρ = constant (uniform distribution)

= variable (non uniform distribution)

where dq is the charge on an infinitesimal volume element dV. Unit of (ρ) is coulomb/meter^3 (C/m^3)

Examples: Charge on a dielectric sphere i.e., non conducting sphere etc.

QUESTION

Which of the following expressions correctly represents the electric field due to a continuous line charge distribution?

A $\vec{E} = \frac{1}{4\pi\varepsilon_0}\sum \frac{q}{r^2}\hat{r}$

B $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda dl}{r^2}\hat{r}$

C $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho dV}{r}\hat{r}$

D $\vec{E} = \lambda r^2$



$\lambda = \frac{dq}{dl} \Rightarrow dq = \lambda dl$

$E = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2}\hat{r}$



QUESTION

Halwaest.

A thin rod of length 1 m carries a uniform charge of 6 μC. What is the linear charge density of the rod?

A. 6 μC/m

B. 3 μC/m

C. 12 μC/m

D. 0.6 μC/m

$$\lambda = \frac{dq}{dl} = \frac{Q}{L}$$

$$\Rightarrow \lambda = \frac{6\mu C}{1m}$$

$$\Rightarrow \boxed{\lambda = 6\mu C/m}$$



QUESTION

A spherical volume has uniform charge density ρ. What is the expression for a small charge element inside it?

A $dq = \rho dr$

B $dq = \rho dA$

C $dq = \rho dV$

D $dq = \lambda dl$

$$\rho = \frac{dq}{dV}$$

$$\Rightarrow \boxed{dq = \rho dV}$$

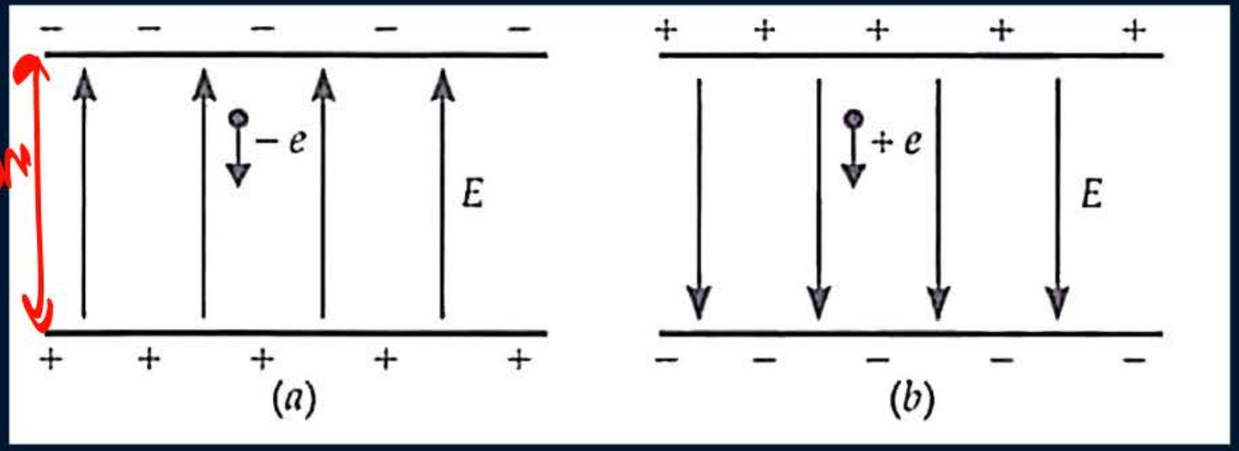


QUESTION

H.W.

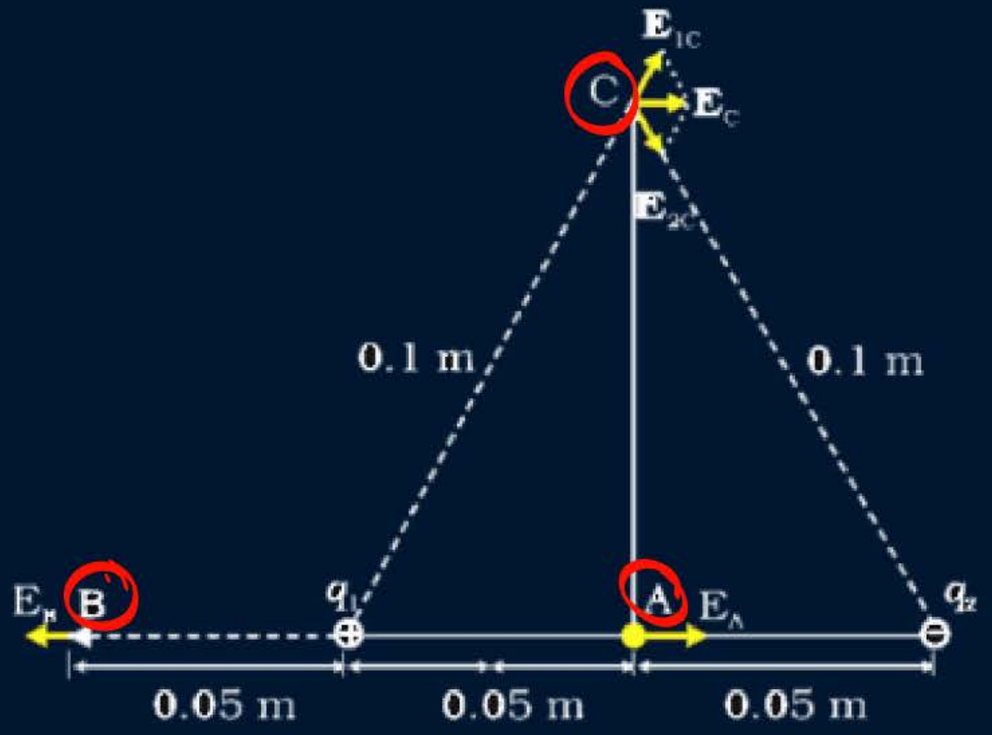
NCERT ex.

An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 NC^{-1} Figure (a). The direction of the field is reversed keeping its magnitude unchanged and a proton falls through the same distance Figure (b). Compute the time of fall in each case. Contrast the situation with that of 'free fall under gravity'.



QUESTION

Two point charges q_1 and q_2. of magnitude -10^{-8} C and -10^{-8} C, respectively, are placed 0.1 m apart. Calculate the electric fields at points A, B and C shown in Figure.



QUESTION

NEET 2022

Hw

Two point charges –q and +q are placed at a distance of L, as shown in the figure
The magnitude of electric field intensity at a distance R (R ≫ L) varies as



A $\frac{1}{R^6}$

B $\frac{1}{R^2}$

C $\frac{1}{R^3}$

D $\frac{1}{R^4}$

$E_{net} = E_A - E_B$

$= \frac{kq}{R^2} - \frac{kq}{(R+L)^2}$

QUESTION

NEET 2023

Two point charges A and B, having charges +Q and –Q respectively, are placed at certain distance apart and force acting between them is F. If 25 % charge of A is transferred to B, then force between the charges becomes

A. F

B. $\frac{9F}{16}$

C. $\frac{16F}{9}$

D. $\frac{4F}{3}$

$$F = \frac{k(+Q)(-Q)}{d^2}$$



$$Q - \frac{25}{100}Q = \frac{3Q}{4}$$

$$-Q + \frac{Q}{4} = \frac{-3Q}{4}$$

$$F' = k\left(\frac{3Q}{4}\right)\left(\frac{3Q}{4}\right)\frac{1}{d^2}$$

$$F' = \frac{9kQQ}{16\,d^2}$$

$$\Rightarrow F' = \frac{9F}{16}$$

THANK YOU

